§3. Passage to finite extension fields

For the classification of number fields and function fields it remains to show how the axioms I through V formulated in the introduction are transferred on passing to finite extension fields.

1. *Suppose that in $\mathfrak{R}$ all axioms I through V are satisfied except axiom II – the multiple-chain condition. Let $\mathfrak{K}$ denote the quotient field of $\mathfrak{R}$, and let $\mathfrak{L}$ be a finite extension field of the first kind over $\mathfrak{K}$.*[14] *Then in the system $\mathfrak{S}$ of all elements of $\mathfrak{L}$ integral with respect to $\mathfrak{R}$ the same axioms are satisfied, and in every order from $\mathfrak{S}$ all these axioms are still satisfied with the exception of integral closedness.*

By §1, 3, $\mathfrak{S}$ forms a ring for which axioms III and IV – existence of the identity element and absence of zero divisors – are satisfied. By the conclusion of §1, 4, $\mathfrak{S}$ is integrally closed in $\mathfrak{L}$; at the same time $\mathfrak{L}$ becomes the quotient field of $\mathfrak{S}$, since every element of $\mathfrak{L}$ is carried into an element of $\mathfrak{S}$ by multiplication with a suitable element of $\mathfrak{R}$. Thus axiom V is satisfied.

The proof of I follows by familiar arguments. As an extension of the first kind, $\mathfrak{L}$ is obtained by adjoining to $\mathfrak{K}$ a single element α, which by the preceding may be assumed integral with respect to $\mathfrak{R}$. Besides α, the conjugate quantities $\alpha', \alpha'', \ldots$ contained in the Galois field of $\mathfrak{L}$ over $\mathfrak{K}$ are also integral; hence so is their product of differences, and so is the square D of this product of differences. Since $\alpha, \alpha', \ldots$ are all distinct, D is a nonzero quantity of $\mathfrak{K}$, hence, by the integral closedness of $\mathfrak{R}$ in $\mathfrak{K}$, an element of $\mathfrak{R}$. By the usual argument, because $\mathfrak{R}$ is integrally closed, $\mathfrak{S}$ is contained in the finite $\mathfrak{R}$-module domain M generated by the elements $\xi_i = \alpha^i/D$; for this one need only pass, in the representation of an element of $\mathfrak{S}$ by the ξ_i, to the conjugate elements and solve the resulting system of linear equations for the coefficients of the representation. By the module theorem of §2, the divisor-chain condition holds for all $\mathfrak{R}$-modules from M, hence in particular for all ideals of $\mathfrak{S}$. The divisor-chain condition likewise holds for the ideals of any order in $\mathfrak{S}$, since here too one is dealing with $\mathfrak{R}$-modules in M; for every order, moreover, axioms III and IV are satisfied.

2. *If all axioms I through V are satisfied in $\mathfrak{R}$, the same is true in $\mathfrak{S}$. In every order from $\mathfrak{S}$ all axioms still hold except integral closedness.*

In view of 1, it remains only to prove axiom II. By the corollary to the module theorem in §2 it is enough to prove the following. If a nonzero ideal of $\mathfrak{S}$ is regarded as an $\mathfrak{R}$-module $\mathfrak{C}$ in M, then the ideals $\mathfrak{c}_i$ of $\mathfrak{R}$ associated with $\mathfrak{C}$ are all different from the zero ideal. Now $\mathfrak{C}$ has the same finite linear rank over $\mathfrak{K}$ as M; for along with any element γ, $\mathfrak{C}$ contains also $\gamma\alpha^i$. Thus for each ξ_i there is an element $c_i \neq 0$ of $\mathfrak{R}$ such that $c_i\xi_i$ belongs to $\mathfrak{C}$; by the uniqueness of the representation by the ξ_i – where the index is to run only through the values less than the degree of the field – c_i belongs to $\mathfrak{c}_i$, and therefore $\mathfrak{c}_i$ is not the zero ideal. This argument remains valid for all orders of the same rank as M; for an order of smaller rank one passes to its quotient field, which as an intermediate field between $\mathfrak{K}$ and $\mathfrak{L}$ is also of the first kind, and whose degree over $\mathfrak{K}$ agrees with the linear rank of the order. The same argument is therefore preserved. Finally, it should be noted that an order in $\mathfrak{S}$ different from $\mathfrak{S}$ is never integrally closed: since every order also contains $\mathfrak{R}$, integral closedness would force it to contain $\mathfrak{S}$.

For the subordination of number fields and function fields, it is therefore enough to verify axioms I through V for the basic domains: the rational integers, polynomials in one indeterminate, and the functional domain of polynomials in several indeterminates.

3. *For the ring $\mathfrak{R}$ of rational integers, respectively of polynomials in one indeterminate with coefficients in a field, axioms I through V are satisfied.* Here I and II follow either from the fact that, modulo any nonzero number or any such polynomial in one indeterminate, there are only finitely many, respectively finitely many linearly independent, residue classes; or from the fact that every ideal is principal. For from this follows the unique decomposition of ideals as a product of powers of prime ideals, and hence a nonzero ideal is divisible by only finitely many ideals. Axioms III and IV are plainly satisfied, the latter as a consequence of the definition of equality. Integral closedness V follows from the fact that, by unique factorization of elements of $\mathfrak{R}$ into irreducibles up to units – divisors of one – every element of the quotient field not contained in $\mathfrak{R}$ can be written as a quotient of two relatively prime elements of $\mathfrak{R}$; hence no power of the numerator is divisible by the denominator. Such an element can therefore satisfy no equation characterizing it as integral with respect to $\mathfrak{R}$.

4. *Let $\mathfrak{R}$ now denote the ring of all polynomials in several indeterminates with coefficients in a field, or with rational integer coefficients.* Then axioms III, IV, V are satisfied as in 3, for here too the factorization of elements into irreducibles is unique up to units. The divisor-chain condition I is an immediate consequence of Hilbert's theorem on the existence of an ideal basis, whereas the multiple-chain condition II does not hold

[14] Following Steinitz, an algebraic extension field is called "of the first kind" if every element is a zero of a prime function which, in a suitable extension field, splits into pairwise distinct linear factors.

here.

However, by passing to the *functional domain*, which amounts to restricting to ideals of highest dimension, one can obtain the validity of axiom II as well; the validity of I can then be proved as in 3, without using Hilbert's theorem. Namely, adjoin to $\mathfrak{R}$ an indeterminate u and consider the ring $\mathfrak{R}^*$ of all polynomials in u with coefficients in $\mathfrak{R}$, equality being defined coefficientwise. As usual, call a polynomial in $\mathfrak{R}^*$ primitive (with respect to $\mathfrak{R}$) if the greatest common divisor of its coefficients – divisor in the sense of a polynomial in $\mathfrak{R}$, not an ideal divisor – is a unit of $\mathfrak{R}$. Then every polynomial in $\mathfrak{R}^*$ is the product of an element of $\mathfrak{R}$ and a primitive polynomial in $\mathfrak{R}^*$, and the product of primitive polynomials in $\mathfrak{R}^*$ is again primitive.

The totality of elements of the quotient field of $\mathfrak{R}^*$ *whose denominators are primitive polynomials in* $\mathfrak{R}^*$ *therefore forms a ring, the functional domain* $\mathfrak{F}$ *of* $\mathfrak{R}$. In this functional domain $\mathfrak{F}$, the units are exactly those elements whose numerator and denominator are primitive polynomials in $\mathfrak{R}^*$; every element of $\mathfrak{F}$ is therefore uniquely representable as the product of an element of $\mathfrak{R}$ and a unit of $\mathfrak{F}$. Hence the factorization theorem for the elements of $\mathfrak{R}$ transfers to the elements of $\mathfrak{F}$; for $\mathfrak{F}$, therefore, in addition to axioms III and IV, axiom V is also satisfied as in 3.

Moreover, in $\mathfrak{F}$ every ideal is principal. For together with any element of $\mathfrak{F}$ an ideal contains also the element of $\mathfrak{R}$ obtained from it by multiplying by a unit; and together with any two elements of $\mathfrak{R}$ it contains their greatest common divisor. Indeed, with $f(x) = t(x)\overline{f}(x)$ and $g(x) = t(x)\overline{g}(x)$ it also contains $t(x)(\overline{f}(x)+u\overline{g}(x))$, and hence $t(x)$, if $\overline{f}$ and $\overline{g}$ are relatively prime and their linear combination is consequently a unit. Thus, starting from an arbitrary element $f(x)$ of the ideal, if there is in the ideal an element $g(x)$ not divisible by it, then the greatest common divisor $t(x)$ also belongs to the ideal and is a proper divisor of $f(x)$. Repeating this finitely many times leads to a basis element of the ideal, and so the ideal is recognized as principal. Thus in $\mathfrak{F}$ the unique decomposition of ideals as products of powers of prime ideals holds, corresponding to the factorization of the elements of $\mathfrak{R}$; as in 3, it follows that axioms I and II are satisfied. The extension fields of the quotient field of $\mathfrak{F}$ that come into consideration here are only those that can be generated by adjoining an element of $\mathfrak{S}$.[15] Taking 1 and 2 into account, this establishes the validity of axioms I through V for number fields and function fields.

§4. Isomorphism theorems. Direct sums

In what follows only the ring property, respectively the module property, is assumed; no further axiom is presupposed.

1. If M and $\overline{M}$ are module domains with respect to $\mathfrak{R}$ (§2), M is called homomorphic to $\overline{M}$ (more precisely, module-homomorphic), $M \sim \overline{M}$, if to every element of M there corresponds one and only one element of $\overline{M}$, in such a way that $\overline{M}$ is exhausted;[16] and if, under this correspondence, difference and multiplication by the same element of $\mathfrak{R}$ correspond. Thus from $\beta \sim \overline{\beta}$ and $\gamma \sim \overline{\gamma}$ it always follows that $(\beta - \gamma) \sim (\overline{\beta} - \overline{\gamma})$ and $r\beta \sim r\overline{\beta}$.

An $\mathfrak{R}$-module $\mathfrak{B}$ in M therefore corresponds homomorphically to an $\mathfrak{R}$-module $\overline{\mathfrak{B}}$ in $\overline{M}$; and the totality $\mathfrak{C}^*$ of elements in M corresponding to elements of an $\mathfrak{R}$-module $\overline{\mathfrak{C}}$ in $\overline{M}$ forms an $\mathfrak{R}$-module in M uniquely determined by $\overline{\mathfrak{C}}$, the module $\mathfrak{C}^*$ associated with $\overline{\mathfrak{C}}$, with $\mathfrak{C}^* \sim \overline{\mathfrak{C}}$. If in particular $\mathfrak{A}$ is the module in M associated with the zero element of $\overline{M}$, then $\mathfrak{C}^*$ is a divisor of $\mathfrak{A}$ and splits into classes of elements of M congruent modulo $\mathfrak{A}$, in such a way that these classes correspond one-to-one to the elements of $\overline{\mathfrak{C}}$. If one passes from $\mathfrak{B}$ in M to $\overline{\mathfrak{B}}$ in $\overline{M}$ and then back to $\mathfrak{B}^*$, one obtains $\mathfrak{B}^* = (\mathfrak{B}, \mathfrak{A})$, the greatest common divisor of $\mathfrak{B}$ and $\mathfrak{A}$; thus $\mathfrak{B}^* = \mathfrak{B}$ if $\mathfrak{B}$ is a divisor of $\mathfrak{A}$.

If the correspondence between the elements of M and $\overline{M}$ is reversible one-to-one, the domains are called isomorphic (more precisely, module-isomorphic), $M \cong \overline{M}$.

If $\mathfrak{A}$ is an arbitrary $\mathfrak{R}$-module in M, then a module domain $\overline{M}$ homomorphic to M arises – the residue-class module $M \mid \mathfrak{A}$ – by taking congruence modulo $\mathfrak{A}$ as the new equality relation. To every element of M there are associated all and only the elements equal to it in $\overline{M}$. Passing in $\overline{M}$ from the equality definition to identity means collecting all equal elements in $\overline{M}$ into one class – a residue class – and taking these residue classes as the new elements of $\overline{M}$.

Every homomorphism is produced by passage to a residue-class module; for if $M \sim \overline{M}$ and if $\mathfrak{A}$ is the module associated with the zero element of $\overline{M}$, then, as shown above, $\overline{M}$ is isomorphic to the residue-class

[15] The system of quantities integral with respect to $\mathfrak{F}$ in these extension fields is also obtained by passing from $\mathfrak{S}$ to a corresponding functional domain, just as one passes from $\mathfrak{R}$ to $\mathfrak{F}$. Compare J. König, *Algebraische Größen*, Leipzig 1903, pp. 468/69.

[16] In the sense of the definition of equality prevailing in $\overline{M}$; compare note 6.

module $M \mid \mathfrak{A}$.

2. **First isomorphism theorem.** *Let $\overline{M}$ be the residue-class module $M \mid \mathfrak{A}$, and let $\mathfrak{C}$ be a divisor of $\mathfrak{A}$. Then there is an isomorphism $\overline{M} \mid \overline{\mathfrak{C}} \cong M \mid \mathfrak{C}$.* Indeed, congruence modulo $\mathfrak{A}$ is at the same time congruence modulo $\mathfrak{C}$; elements equal modulo $\mathfrak{A}$ remain equal modulo $\mathfrak{C}$. Thus one may form the residue-class module $M \mid \mathfrak{C}$ – that is, equality modulo $\mathfrak{C}$ – by first identifying elements modulo $\mathfrak{A}$, passing to $\overline{M}$, and then collecting in $\overline{M}$ those elements that are equal modulo $\mathfrak{C}$; this is the same as identifying modulo $\overline{\mathfrak{C}}$ in $\overline{M}$, i.e. forming $\overline{M} \mid \overline{\mathfrak{C}}$.

Second isomorphism theorem. *If $\mathfrak{B}$ and $\mathfrak{A}$ are modules in M, then $(\mathfrak{B}, \mathfrak{A}) \mid \mathfrak{A} \cong \mathfrak{B} \mid [\mathfrak{B}, \mathfrak{A}]$.* For by 1, $\mathfrak{B}$ becomes homomorphic to $\overline{\mathfrak{B}}$ when $(\mathfrak{B}, \mathfrak{A}) \mid \mathfrak{A}$ is put equal to $\overline{\mathfrak{B}}$; and since precisely the elements of $[\mathfrak{B}, \mathfrak{A}]$ correspond to the zero element in $\overline{\mathfrak{B}}$, the asserted isomorphism again follows from 1.

3. If $\mathfrak{R}$ and $\overline{\mathfrak{R}}$ are (commutative) rings, $\mathfrak{R}$ is called homomorphic to $\overline{\mathfrak{R}}$ (more precisely, ring-homomorphic), $\mathfrak{R} \sim \overline{\mathfrak{R}}$, if to each element of $\mathfrak{R}$ there corresponds one and only one element of $\overline{\mathfrak{R}}$ in such a way that $\overline{\mathfrak{R}}$ is exhausted, and if differences and products correspond under this assignment. If the correspondence of elements is reversible one-to-one, the rings are called isomorphic (more precisely, ring-isomorphic), $\mathfrak{R} \cong \overline{\mathfrak{R}}$.

All the arguments in 1 and 2 remain valid when the modules are replaced by ideals in $\mathfrak{R}$,[17] and when module-homomorphism and module-isomorphism are replaced by ring-homomorphism and ring-isomorphism. In particular, if $\mathfrak{c}$ is a divisor of $\mathfrak{a}$, the residue-class ring $\mathfrak{c} \mid \mathfrak{a}$ is obtained by introducing congruence modulo $\mathfrak{a}$ as the new equality relation; here one must note that products of residue classes are now also defined.

Thus $\mathfrak{c}$ is homomorphic to $\mathfrak{c} \mid \mathfrak{a}$; every homomorphism is produced in this way, and the isomorphism theorems hold as in 2:

First isomorphism theorem. If $\overline{\mathfrak{R}}$ denotes the residue-class ring $\mathfrak{R} \mid \mathfrak{a}$, and $\mathfrak{c}$ is a divisor of $\mathfrak{a}$, then $\overline{\mathfrak{R}} \mid \bar{\mathfrak{c}} \cong \mathfrak{R} \mid \mathfrak{c}$ in the sense of ring isomorphism.

Second isomorphism theorem. If $\mathfrak{b}$ and $\mathfrak{a}$ are ideals of $\mathfrak{R}$, then the ring-isomorphism $(\mathfrak{b}, \mathfrak{a}) \mid \mathfrak{a} \cong \mathfrak{b} \mid [\mathfrak{b}, \mathfrak{a}]$ holds.[18]

4. I now collect the calculation rules for coprime ideals[19] from which the theorems on direct sums in 5 follow. In the commutative ring $\mathfrak{R}$, assume the existence of a multiplicative identity element e. Thus $\mathfrak{a} = \mathfrak{o}\mathfrak{a}$ for every ideal of $\mathfrak{R}$, where $\mathfrak{o}$ denotes the unit ideal. Two ideals $\mathfrak{a}, \mathfrak{b}$ are called coprime if their greatest common divisor $(\mathfrak{a}, \mathfrak{b})$ is $\mathfrak{o}$.

4α. If $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$, then $(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = (\mathfrak{a}, \mathfrak{c})$. For $(\mathfrak{a}, \mathfrak{c}) = (\mathfrak{a}, \mathfrak{b})(\mathfrak{a}, \mathfrak{c}) = (\mathfrak{a}^2, \mathfrak{a}\mathfrak{b}, \mathfrak{a}\mathfrak{c}, \mathfrak{b}\mathfrak{c})$ is divisible by $(\mathfrak{a}, \mathfrak{b}\mathfrak{c})$ and conversely. Hence:

4β. From $\mathfrak{c}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ and $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ it follows that $\mathfrak{c} \equiv 0 \pmod{\mathfrak{a}}$. Indeed, $\mathfrak{c}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ is equivalent to $(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = \mathfrak{a}$; because $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$, this also gives $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{a}$, or $\mathfrak{c} \equiv 0 \pmod{\mathfrak{a}}$. Furthermore:

4γ. If each of the ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$ is coprime to each of the ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_s$, then the product of the $\mathfrak{a}_i$ is coprime to the product of the $\mathfrak{b}_i$. For from $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ and $(\mathfrak{a}, \mathfrak{c}) = \mathfrak{o}$ one obtains $(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = \mathfrak{o}$, and finite repetition gives the assertion.

From 4α and 4γ follows:

4δ. If the ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_r$ are pairwise coprime, i.e. $(\mathfrak{b}_i, \mathfrak{b}_k) = \mathfrak{o}$ for $i \neq k$, then their least common multiple equals their product. Let $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ and $\mathfrak{v} = [\mathfrak{a}, \mathfrak{b}]$. Then $\mathfrak{v} = \mathfrak{o}\mathfrak{v} = (\mathfrak{a}\mathfrak{v}, \mathfrak{b}\mathfrak{v}) = 0(\mathfrak{a}, \mathfrak{b})$; since $\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{v}}$, it follows that $\mathfrak{v} = \mathfrak{a}\mathfrak{b}$. Finite repetition, using 4γ, gives the claim.

4ε. If the ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_r$ are pairwise coprime and $\mathfrak{a}_i = [\mathfrak{b}_1, \ldots, \mathfrak{b}_{i-1}, \mathfrak{b}_{i+1}, \ldots, \mathfrak{b}_r] = \mathfrak{b}_1 \cdots \mathfrak{b}_{i-1}\mathfrak{b}_{i+1} \cdots \mathfrak{b}_r$, then $(\mathfrak{a}_1, \ldots, \mathfrak{a}_{i-1}, \mathfrak{a}_{i+1}, \ldots, \mathfrak{a}_r) = \mathfrak{b}_i$, and hence $(\mathfrak{a}_1, \ldots, \mathfrak{a}_r) = \mathfrak{o}$. For by the distributive law $(\mathfrak{a}_1, \mathfrak{a}_2) = \mathfrak{b}_3 \cdots \mathfrak{b}_r(\mathfrak{b}_2, \mathfrak{b}_1) = \mathfrak{b}_3 \cdots \mathfrak{b}_r$; hence $(\mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3) = \mathfrak{b}_4 \cdots \mathfrak{b}_r(\mathfrak{b}_3, \mathfrak{b}_1\mathfrak{b}_2) = \mathfrak{b}_4 \cdots \mathfrak{b}_r$, and finite repetition, after a suitable numbering, proves the assertion.

The ideal $\mathfrak{a}_i$ is called the complement of $\mathfrak{b}_i$ in the representation $\mathfrak{m} = [\mathfrak{b}_1, \ldots, \mathfrak{b}_r]$.

5. Again let the commutative ring $\mathfrak{R}$ have a multiplicative identity element. The ring $\mathfrak{R}$ is called the direct sum of the ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$, in symbols $\mathfrak{R} = \mathfrak{a}_1 + \mathfrak{a}_2 + \cdots + \mathfrak{a}_r$, if each element c of $\mathfrak{R}$ can be represented in one and only one way in the form $c = a_1 + a_2 + \cdots + a_r$, with a_i an element of $\mathfrak{a}_i$.

If the zero ideal is the least common multiple of the pairwise coprime ideals $\mathfrak{b}_1, \ldots, \mathfrak{b}_r$, and if $\mathfrak{a}_i$ is the complement of $\mathfrak{b}_i$, then $\mathfrak{R}$ is the direct sum of the ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_r$. Since $\mathfrak{o} = (\mathfrak{a}_1, \ldots, \mathfrak{a}_r)$, every element of $\mathfrak{R}$ admits at least one additive representation by the $\mathfrak{a}_i$. Since $[\mathfrak{a}_i, \mathfrak{b}_i] = (0)$, this representation is unique: from

[17] For noncommutative rings, two-sided ideals must be used.

[18] These isomorphism theorems, familiar from group theory, occur for modules in a somewhat more special form first in Dedekind (compare 3rd ed., p. 484). The above form for ideals occurs in Masazo Sono, *On Congruences. I, II, III, IV*, Memoirs of the College of Science, Kyoto Imperial University, 2 (1917), 3 (1918, 1919), compare 2, p. 215. In each case only the second isomorphism theorem is stated explicitly.

[19] In the form going back to Dedekind (4th ed., §178, III to VIII). The calculation with the identity element that appears throughout more recent accounts is much more cumbersome.

$0 = a_1 + a_2 + \cdots + a_r = a_i + b_i$ one obtains $a_i = 0$. By the second isomorphism theorem the $\mathfrak{a}_i$ are isomorphic to the residue-class rings $\mathfrak{R} \mid \mathfrak{b}_i$. The orthogonality relations $\mathfrak{a}_i\mathfrak{a}_k = (0)$ for $i \neq k$, and $\mathfrak{a}_i^2 = \mathfrak{a}_i$, hold; the latter because $\mathfrak{a}_i = \mathfrak{o}\mathfrak{a}_i$.

Conversely, if there is a representation of $\mathfrak{R}$ *as a direct sum,* $\mathfrak{R} = \mathfrak{a}_1 + \mathfrak{a}_2 + \cdots + \mathfrak{a}_r$*, and if one sets* $\mathfrak{b}_i = (\mathfrak{a}_1, \ldots, \mathfrak{a}_{i-1}, \mathfrak{a}_{i+1}, \ldots, \mathfrak{a}_r) = \mathfrak{a}_1 + \cdots + \mathfrak{a}_{i-1} + \mathfrak{a}_{i+1} + \cdots + \mathfrak{a}_r$*, then the* $\mathfrak{b}_i$ *are pairwise coprime and their least common multiple is the zero ideal.* For from $\mathfrak{o} = (\mathfrak{a}_i, \mathfrak{b}_i)$ follows $\mathfrak{o} = (\mathfrak{b}_k, \mathfrak{b}_i)$ for $k \neq i$, since $\mathfrak{b}_k$ is a divisor of $\mathfrak{a}_i$. If, moreover, c is divisible by all $\mathfrak{b}_i$, then

$$c = a_2^{(1)} + \cdots + a_r^{(1)} = a_1^{(2)} + a_3^{(2)} + \cdots + a_r^{(2)} = \cdots = a_1^{(r)} + \cdots + a_{r-1}^{(r)},$$

where in each case $a_i^{(\lambda)} \equiv 0 \pmod{\mathfrak{a}_i}$. By the uniqueness of the representation, since in each representation one component is zero, it follows that $0 = a_1^{(\lambda)}, \ldots, 0 = a_r^{(\lambda)}$ for every λ, and hence $c = 0$.[20]

§5. Prime ideals and primary ideals

Let $\mathfrak{R}$ again be a commutative ring for which no further axiom, not even the existence of an identity element, is assumed. We collect the basic facts about prime ideals and primary ideals.[21]

1. An ideal $\mathfrak{p}$ of $\mathfrak{R}$ is called a *weak prime ideal* if the residue-class ring $\mathfrak{R} \mid \mathfrak{p}$ is a ring without zero divisors; that is, if from $a \not\equiv 0 \pmod{\mathfrak{p}}$ and $b \not\equiv 0 \pmod{\mathfrak{p}}$ it always follows that $ab \not\equiv 0 \pmod{\mathfrak{p}}$. An ideal $\mathfrak{p}^*$ of $\mathfrak{R}$ is called a *strong prime ideal* if the residue-class ring $\mathfrak{R} \mid \mathfrak{p}^*$ is a ring without zero-divisor ideals; that is, if from $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}^*}$ and $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}^*}$ it always follows that $\mathfrak{a}\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}^*}$.

Every strong prime ideal is at the same time a weak prime ideal, as the specialization of $\mathfrak{a}, \mathfrak{b}$ to principal ideals shows. Conversely, *every weak prime ideal is also a strong prime ideal*; one can therefore simply speak of prime ideals. For let $\mathfrak{p}$ be a weak prime ideal; let $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}}$ and $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$; choose elements a and b in $\mathfrak{a}$ and $\mathfrak{b}$ such that $a \not\equiv 0 \pmod{\mathfrak{p}}$ and $b \not\equiv 0 \pmod{\mathfrak{p}}$. Then $ab \not\equiv 0 \pmod{\mathfrak{p}}$, hence $\mathfrak{a}\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$, so $\mathfrak{p}$ is also a strong prime ideal.

2. An ideal $\mathfrak{q}$ of $\mathfrak{R}$ is called a *weak primary ideal* if in the residue-class ring $\mathfrak{R} \mid \mathfrak{q}$ some power of every zero divisor vanishes; equivalently, if from $a \not\equiv 0 \pmod{\mathfrak{q}}$ and $b^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every x it always follows that $ab \not\equiv 0 \pmod{\mathfrak{q}}$. The system $\mathfrak{p}$ of all elements of $\mathfrak{R}$ that become zero divisors in $\mathfrak{R} \mid \mathfrak{q}$ forms an ideal, indeed a prime ideal; it is a divisor of $\mathfrak{q}$ and is called the associated prime ideal. For along with a, also ra is a zero divisor; along with a and b, so is $a - b$, since with a^x and b^λ the element $(a-b)^{x+\lambda}$ is always divisible by $\mathfrak{q}$. If furthermore a is not a zero divisor, then by definition no power of a is a zero divisor; from $a \not\equiv 0 \pmod{\mathfrak{p}}$ and $b \not\equiv 0 \pmod{\mathfrak{p}}$ follows $a^x b^\lambda = (ab)^x \not\equiv 0 \pmod{\mathfrak{q}}$, hence $ab \not\equiv 0 \pmod{\mathfrak{p}}$.

An ideal $\mathfrak{q}^*$ of $\mathfrak{R}$ is called a *strong primary ideal* if in the residue-class ring $\mathfrak{R} \mid \mathfrak{q}^*$ some power of every zero-divisor ideal vanishes; that is, if from $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{q}^*}$ and $\mathfrak{b}^x \not\equiv 0 \pmod{\mathfrak{q}^*}$ for every x it always follows that $\mathfrak{a}\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{q}^*}$. As for weak primary ideals one shows: the greatest common divisor of all ideals of $\mathfrak{R}$ that become zero-divisor ideals in $\mathfrak{R} \mid \mathfrak{q}^*$ forms a prime ideal, a divisor of $\mathfrak{q}^*$, the associated prime ideal. Conversely, all primary ideals with the same associated prime ideal $\mathfrak{p}$ are said to belong to $\mathfrak{p}$.

Every strong primary ideal is at the same time a weak primary ideal, as the specialization of $\mathfrak{a}, \mathfrak{b}$ to principal ideals shows. In general, however, the converse is false.[22] *If, however, axiom I, the divisor-chain condition, is assumed in* $\mathfrak{R}$*, then every weak primary ideal is also a strong primary ideal;* one can therefore simply speak of primary ideals. Let $\mathfrak{q}$ be a weak primary ideal; suppose $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{q}}$ and $\mathfrak{b}^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every x. Then there exist elements a of $\mathfrak{a}$ and b of $\mathfrak{b}$ such that $a \not\equiv 0 \pmod{\mathfrak{q}}$ and $b^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every x; hence $ab \not\equiv 0 \pmod{\mathfrak{q}}$ and therefore $\mathfrak{a}\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{q}}$. For if to every b in $\mathfrak{b}$ there were an exponent λ such that $b^\lambda \equiv 0 \pmod{\mathfrak{q}}$, then $\mathfrak{b}^{\lambda_1 + \cdots + \lambda_r} \equiv 0 \pmod{\mathfrak{q}}$, with $\lambda_1, \ldots, \lambda_r$ the exponents of the finitely many basis elements.[23] The same argument also shows that there is a smallest exponent ϱ such that $\mathfrak{p}^\varrho \equiv 0 \pmod{\mathfrak{q}}$, where $\mathfrak{p}$ is the associated prime ideal; ϱ is called the exponent of $\mathfrak{q}$.

[20] The theorems given under 5 are special cases of a general theorem on the connection between direct sum and direct intersection. Compare H. Prüfer, Theorie der Abelschen Gruppen I, Math. Zeitschr. 20 (1924), pp. 165–187, §6. The arguments given there remain valid also for such a general formulation of the concept of group that modules and ideals fall under it as special cases.

[21] In *Idealtheorie*, §4, axiom I, the divisor-chain condition, is assumed; therefore it is not made sharp there which properties of prime and primary ideals are independent of this axiom.

[22] This is shown by the following example communicated to me by R. Hölzer. Let $\mathfrak{R}$ be the polynomial domain in countably many indeterminates x_i with coefficients in a field, and let $\mathfrak{q} = (x_1^2, x_2^3, \ldots, x_\nu^{\nu+1}, \ldots, x_i x_k\ [i \neq k] \ldots)$. The zero divisors in the residue-class ring are exactly the polynomials divisible by $\mathfrak{p} = (x_1, x_2, \ldots, x_\nu, \ldots)$, and in each case a power of such a zero divisor is divisible by $\mathfrak{q}$; therefore $\mathfrak{q}$ is a weak primary ideal with associated prime ideal $\mathfrak{p}$. But $\mathfrak{q}$ is not a strong primary ideal: putting $\mathfrak{a} = (x_1, x_3, x_5, \ldots, x_{2\nu+1}, \ldots)$ and $\mathfrak{b} = (x_2, x_4, \ldots, x_{2\nu}, \ldots)$, one has $\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{q}}$, but no power of $\mathfrak{a}$ or $\mathfrak{b}$ is divisible by $\mathfrak{q}$.

[23] In order to infer the finite ideal basis from axiom I, a fixed well-ordering in $\mathfrak{R}$ must be used; compare §6.

3. In what follows the divisor-chain condition is again *not* assumed. The least common multiple $[\mathfrak{a}_1, \dots, \mathfrak{a}_r]$ of finitely many ideals is called a *shortest representation* if no $\mathfrak{a}_i$ is contained in the least common multiple of the remaining ideals, i.e. if no $\mathfrak{a}_i$ can be omitted.

The least common multiple of finitely many weak, respectively strong, primary ideals belonging to the same prime ideal $\mathfrak{p}$ *is again a weak primary ideal with* $\mathfrak{p}$ *as associated prime ideal. A shortest representation by finitely many weak, respectively strong, primary ideals belonging to distinct prime ideals is not a weak, respectively strong, primary ideal.* Let $\mathfrak{f} = [\mathfrak{q}_1, \dots, \mathfrak{q}_r]$, and let $\mathfrak{p}$ be the associated prime ideal of the weak primary ideals $\mathfrak{q}_i$. Then $\mathfrak{p}$ also consists exactly of the elements of which some power is divisible by $\mathfrak{f}$. From $a \not\equiv 0 \pmod{\mathfrak{f}}$ and $b^x \not\equiv 0 \pmod{\mathfrak{f}}$ for every x it follows that $b \not\equiv 0 \pmod{\mathfrak{p}}$ and $a \not\equiv 0 \pmod{\mathfrak{q}_i}$ for at least one index i; hence $ab \not\equiv 0 \pmod{\mathfrak{q}_i}$, and therefore $ab \not\equiv 0 \pmod{\mathfrak{f}}$. If one replaces elements throughout by ideals, one can no longer infer $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$.

Now let $\mathfrak{m} = [\mathfrak{q}_1, \dots, \mathfrak{q}_r]$ be a shortest representation by weak primary ideals with $\mathfrak{p}_i \neq \mathfrak{p}_k$ for $i \neq k$, and let $r \geq 2$. There is then at least one associated prime ideal, say $\mathfrak{p}_1$, which is not contained in any of the others; for every proper multiple-chain of the $\mathfrak{p}$'s must terminate after at most r steps. Hence there are elements a_i of $\mathfrak{p}_i$ such that $a_i^{\varrho_i} \equiv 0 \pmod{\mathfrak{q}_i}$, while $a_2 \not\equiv 0 \pmod{\mathfrak{p}_1}, \dots, a_r \not\equiv 0 \pmod{\mathfrak{p}_1}$. If further $q_1 \not\equiv 0 \pmod{\mathfrak{m}}$ is an element of $\mathfrak{q}_1$, then $q_1 a_2^{\varrho_2} \cdots a_r^{\varrho_r} \equiv 0 \pmod{\mathfrak{m}}$, but no power of $a_2^{\varrho_2} \cdots a_r^{\varrho_r}$ is divisible by $\mathfrak{m}$, since otherwise divisibility by $\mathfrak{p}_1$ would follow. Thus $\mathfrak{m}$ is not a weak primary ideal. If elements are replaced throughout by ideals, that is q_1 by $\mathfrak{q}_1$ and a_i by $\mathfrak{a}_i[\not\equiv 0 \pmod{\mathfrak{p}_1}$, but $\mathfrak{a}_i^{\varrho_i} \equiv 0 \pmod{\mathfrak{q}_i}]$, the proof for strong primary ideals is obtained.[24]

4. *Module quotient and ideal quotient.* Let $\mathfrak{T}$ be an extension ring of $\mathfrak{R}$, and let $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \dots$ be $\mathfrak{R}$-modules in $\mathfrak{T}$. The quotient $\mathfrak{C} = \mathfrak{A} : \mathfrak{B}$ is defined as a module satisfying the conditions $\mathfrak{C}\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$ and, if $\mathfrak{D}\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$, then $\mathfrak{D} \equiv 0 \pmod{\mathfrak{C}}$; thus $\mathfrak{C}$ is the smallest module for which $\mathfrak{C}\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$ holds. This determines the quotient uniquely as the greatest common divisor of all modules $\mathfrak{D}$ for which $\mathfrak{D}\mathfrak{B} \equiv 0 \pmod{\mathfrak{A}}$; such $\mathfrak{D}$ exist, since the zero module is certainly one of them.

From the definition it follows: if $\mathfrak{B}$ is a multiple of $\overline{\mathfrak{B}}$, then $\mathfrak{A} : \overline{\mathfrak{B}}$ is a multiple of $\mathfrak{A} : \mathfrak{B}$. If $\mathfrak{A}$ is a multiple of $\overline{\mathfrak{A}}$, then $\mathfrak{A} : \mathfrak{B}$ is a multiple of $\overline{\mathfrak{A}} : \mathfrak{B}$. Also

$$\mathfrak{A} : \mathfrak{B}\mathfrak{C} = (\mathfrak{A} : \mathfrak{B}) : \mathfrak{C} = (\mathfrak{A} : \mathfrak{C}) : \mathfrak{B}.$$

If the extension ring $\mathfrak{T}$ coincides with $\mathfrak{R}$, the module quotient becomes the ideal quotient $\mathfrak{a} : \mathfrak{b}$; hence the preceding rules also hold here. Since $\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$, one also has $\mathfrak{a} \equiv 0 \pmod{\mathfrak{a} : \mathfrak{b}}$.

5. *If* $\mathfrak{a} : \mathfrak{b} = \mathfrak{a}$, *then* $\mathfrak{b}$ *is called prime to* $\mathfrak{a}$. Thus an ideal $\mathfrak{b}$ is prime to $\mathfrak{a}$ if and only if $\mathfrak{d}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ always implies $\mathfrak{d} \equiv 0 \pmod{\mathfrak{a}}$. From the calculation rules under 4 it follows: if $\mathfrak{b}$ and $\mathfrak{c}$ are prime to $\mathfrak{a}$, then so are their product and their least common multiple. If $\mathfrak{b}$ is prime to $\mathfrak{a}$ and $\mathfrak{a}$ is prime to $\mathfrak{b}$, then $\mathfrak{a}$ and $\mathfrak{b}$ are called mutually prime. From §4, 4β, it follows that coprime ideals are also mutually prime.

[24]I owe this modification of my original proof, which makes the divisor-chain condition unnecessary, to B. L. van der Waerden.